

ABM BUSINESS LOGIC BIBLE

SIGNAL SOURCE COVERAGE GAP ANALYSIS

OPEN-GAP-SIG Series | 7 Gaps Identified | Governance Decisions Required

CONTEXT & PURPOSE

This document formalises the signal source coverage gaps identified during the INT-SIG exhaustiveness review. The V2 spec defines eight intelligence sub-streams (SIG-NEWS, SIG-EVENT, SIG-FIN, SIG-REG, SIG-EXEC, SIG-VENDOR, SIG-SUBS, SIG-PATH) plus four enrichment providers (Apollo, Clearbit, Clay, LinkedIn). This set is internally coherent but not externally complete: several high-value signal categories that the V1.3 FSD references or that the KSA banking domain demands are absent from V2's harvester architecture.

Each gap below follows the project's established pattern: the failure it prevents is identified, the root cause is stated, candidate source layers are listed, and a governance decision is required. These are OPEN-GAP items with the same first-class status as OPEN-Q items: safe defaults exist (the engine operates without them), but the absence represents a known blind spot that degrades scoring accuracy, timing sensitivity, and competitive intelligence completeness.

GAP SUMMARY

Gap ID	Missing Stream	Severity	V1.3 Mention?	Governance Decision
OPEN-GAP-SIG-01	SIG-HIRE (Hiring/Job Postings)	CRITICAL	Yes — listed in architecture, enrichment	Add as 9th sub-stream or fold into SIG-EXEC?
OPEN-GAP-SIG-02	SIG-TENDER (Procurement/RFP)	CRITICAL	No	Add as 10th sub-stream? Source availability in KSA?
OPEN-GAP-SIG-03	SIG-FUND (Funding Rounds)	HIGH	Yes — V1.3 architecture + source reliability	Add as sub-stream or fold into SIG-FIN?
OPEN-GAP-SIG-04	SIG-SOCIAL (Twitter/X)	MEDIUM	No	Add as source layer under SIG-EXEC + SIG-NEWS?
OPEN-GAP-SIG-05	SIG-ANALYST (Industry Reports)	MEDIUM	No	Add as source under EXEC_BELIEF_TIMELINE?
OPEN-GAP-SIG-06	SIG-PARTNER (MOU/Partnership)	HIGH	No	Fold into SIG-VENDOR or separate stream?
OPEN-GAP-SIG-07	SIG-IP (Patent/Trademark)	LOW	No	Defer to P2 or drop?

DETAILED GAP SPECIFICATIONS

OPEN-GAP-SIG-01: SIG-HIRE — Hiring & Job Posting Intent Signals

CRITICAL

V1.3 FSD Reference: V1.3 Section 2.1 lists 'Hiring' as an external signal category. V1.3 Section 5.3 lists 'Analyze Hiring Trends' as an enrichment workflow step. V1.3 Section 3.1.1 references hiring signals. Funding DBs rated 0.80 in ICS-SOURCE-001. Yet V2 INT-SIG has no SIG-HIRE harvester.

FAILURE PREVENTED: A Saudi bank posting 'Head of Digital Lending Platform' or 'Core Banking Migration Lead' on LinkedIn Jobs, Bayt, or their careers page is one of the highest-confidence buying-intent signals in enterprise fintech. Without SIG-HIRE, the engine is blind to explicit demand signals that directly map to Decimal's product portfolio. The bank is literally advertising that it needs what Decimal sells, and the engine cannot see it.

Root Cause: V1.3 mentioned hiring signals but V2's INT-SIG rewrite (which was built from the 5-Whys methodology) focused on the eight streams that emerged from the SNB/Riyad/Al-Rajhi manual BD methodology. Hiring signal harvesting was not part of that manual process (Decimal's BD team used hiring as a mental heuristic, not a systematic source), so it never surfaced in the 5-Whys excavation.

Candidate Source Layer	LinkedIn Jobs API (or scrape via Heyreach), Bayt.com, GulfTalent, bank career pages (scrape), Jadara (Saudi job platform)
Signal Meaning	Buying intent — budget committed, role scoped, technology direction revealed. A job posting is a procurement signal that precedes the RFP by 6-18 months.
Scoring Trigger Example	Bank posts 'Digital Lending Transformation Lead' with Temenos/Backbase/FIS in requirements -> priority bump + SIG-VENDOR cross-link (incumbent tech stack revealed) + timing window (hiring = 3-6 month implementation horizon)
Cadence	Daily (LinkedIn Jobs, Bayt) + event-driven (bank career page scrape weekly)
Existing Section Binding	Feeds M1 (signal), M2 (scoring via Intent Component), SIG-VENDOR (tech stack inference from job requirements), SIG-EXEC (new-hire detection as executive-move signal), Timing module (hiring-to-procurement lag model)
Phase Recommendation	P1 — This is the single highest-value gap. LinkedIn Jobs is accessible via Heyreach or API. Minimal source integration, maximum signal value.

GOVERNANCE DECISION REQUIRED: Should SIG-HIRE be a standalone 9th sub-stream with its own 5-Whys, or should it be folded into SIG-EXEC as a sub-harvester? Argument for standalone: different source layer, different cadence, different signal meaning (buying intent vs. executive dossier). Argument for fold-in: executive hiring IS an exec-move event. Recommendation: standalone — the signal meaning (demand signal) is fundamentally different from SIG-EXEC's purpose (who to reach).

SAFE DEFAULT (if unresolved): Engine operates without SIG-HIRE. Hiring-intent signals are invisible; scoring relies on the remaining seven signal types. No false positives, but systematic false negatives on accounts in active procurement preparation.

OPEN-GAP-SIG-02: SIG-TENDER — Procurement & RFP Signals

CRITICAL

V1.3 FSD Reference: Not referenced in V1.3 or V2.

FAILURE PREVENTED: Saudi banks and government-adjacent entities publish tenders and RFPs through Etimad (government procurement platform), direct procurement portals, and occasionally through consulting intermediaries. An active tender for 'Digital Lending Platform' or 'Customer Onboarding System' is the highest-possible-confidence buying signal — the decision to buy has already been made; the question is from whom. Missing this means the engine discovers opportunities only through indirect signals when the direct signal was publicly available.

Root Cause: The manual BD methodology that seeded INT-SIG did not include systematic tender monitoring (Decimal's BD team relied on network intelligence for RFP awareness). The 5-Whys excavation therefore never surfaced procurement signals as a stream.

Candidate Source Layer	Etimad (etimad.sa — Saudi government procurement), bank procurement portals (where public), MENA procurement aggregators, consulting-firm RFP distribution lists (Deloitte, EY, McKinsey — when acting as procurement advisors)
Signal Meaning	Demand confirmation — budget allocated, scope defined, timeline committed. The buying decision is made; the vendor selection is in progress.
Scoring Trigger Example	Bank publishes RFP for 'Digital Lending Platform Modernization' on Etimad -> CRITICAL priority + opportunity state machine transition to ACTIVE PROCUREMENT + all outreach shifted to proposal-support mode
Cadence	Daily (Etimad RSS/scrape) + event-driven (procurement portal monitoring)
Existing Section Binding	Feeds M2 (scoring — highest possible intent weight), Opportunity State Machine (\$14 — triggers ACTIVE PROCUREMENT state), Timing module (RFP deadline = hard window), SIG-VENDOR (incumbent revealed in tender scope), \$5 Resource Allocation (tender-response = highest priority resource commitment)
Phase Recommendation	P1 (manual input) / P2 (automated) — Manual tender input is trivially implementable (human enters known RFP into Signal entity). Automated monitoring depends on source availability assessment.

GOVERNANCE DECISION REQUIRED: Is systematic tender monitoring feasible in KSA? Etimad covers government and quasi-government entities but not all private-sector bank procurement. If source availability is limited, should SIG-TENDER be a manual-input stream (BD team feeds known tenders) with automation added in P2? Or is systematic scraping viable now?

SAFE DEFAULT (if unresolved): Engine operates without SIG-TENDER. Active procurement opportunities are discovered only through network intelligence or indirect signals. No false positives, but the highest-confidence buying signals are invisible to the automated engine.

OPEN-GAP-SIG-03: SIG-FUND — Funding & Investment Signals**HIGH**

V1.3 FSD Reference: V1.3 Section 2.1 lists 'Funding' as an external signal. ICS-SOURCE-001 assigns funding databases a 0.80 source reliability rating. V2's SIG-FIN covers financial disclosures (quarterly results, annual reports) but not venture/strategic investment rounds.

FAILURE PREVENTED: Saudi bank subsidiaries (NEO, Emkan, STC Pay partnerships) and fintech ventures raise funding rounds that signal technology investment direction and budget availability. A Series B for a bank's digital subsidiary indicates committed infrastructure spend. Without SIG-FUND, the engine conflates parent-bank financial health (SIG-FIN) with subsidiary investment activity — they are different signals with different meanings.

Root Cause: SIG-FIN was scoped to public financial disclosures (Tadawul, IR pages, rating agencies). Venture funding is a different source layer (Crunchbase, MAGNiTT, Wamda) with different entities (subsidiaries, not parent banks). The V2 5-Whys for SIG-FIN stopped at the parent-bank level.

Candidate Source Layer	Crunchbase API, MAGNiTT (MENA startup database), Wamda, bank press releases (parsed for investment announcements), Saudi Venture Capital Company (SVC) disclosures
Signal Meaning	Budget availability + technology direction at the subsidiary/venture level. Distinct from parent financial health.
Scoring Trigger Example	Bank subsidiary raises \$50M Series B for digital banking platform -> budget-availability signal + technology-direction signal + subsidiary priority bump
Cadence	On-publish + weekly scan
Existing Section Binding	Feeds M2 (scoring via Budget Modifier in Effective Opportunity), SIG-SUBS (subsidiary activity), SIG-FIN (complements but does not duplicate — different entity level)
Phase Recommendation	P1 — MAGNiTT and Crunchbase APIs are accessible. KSA fintech funding is actively tracked by both platforms. Integration complexity is low.

GOVERNANCE DECISION REQUIRED: Should SIG-FUND be a standalone sub-stream or folded into SIG-FIN with expanded scope? Argument for fold-in: both are financial signals about budget availability. Argument for standalone: different source layer (venture DBs vs. Tadawul), different entity level (subsidiary vs. parent), different signal meaning (investment direction vs. financial health).

SAFE DEFAULT (if unresolved): Engine operates without SIG-FUND. Subsidiary investment activity is captured only if it surfaces in SIG-NEWS (press coverage). Scoring treats parent financial health as proxy for subsidiary budget — inaccurate but not dangerous.

OPEN-GAP-SIG-04: SIG-SOCIAL — Twitter/X & Social Media Signals**MEDIUM**

V1.3 FSD Reference: Not referenced in V1.3 or V2.

FAILURE PREVENTED: Saudi banking executives are unusually active on Twitter/X compared to other markets. CxO commentary on digital transformation, open banking, or fintech partnerships surfaces belief trajectories (EXEC_BELIEF_TIMELINE) and relationship signals (who engages with whom) that LinkedIn alone does not capture. The KSA Twitter ecosystem is a real-time sentiment and priority signal source.

Root Cause: The manual BD methodology was LinkedIn-centric. Twitter/X was not a systematic source in the SNB/Riyad/Al-Rajhi intelligence process, so it never entered the 5-Whys.

Candidate Source Layer	Twitter/X API (or scrape), executive account monitoring, hashtag tracking (#SaudiFintech, #OpenBanking, #Vision2030Fintech)
Signal Meaning	Belief trajectory + sentiment + relationship mapping. Complements SIG-EXEC (speeches/quotes) with real-time, unfiltered executive commentary.
Scoring Trigger Example	Bank CTO tweets thread on open banking challenges -> EXEC_BELIEF_TIMELINE update + personalization material (grounded, sourced) + timing signal (topic is top-of-mind now)
Cadence	Continuous (monitored accounts) + daily (hashtag/keyword scan)

Existing Section Binding	Feeds SIG-EXEC (additional grounded material), EXEC_BELIEF_TIMELINE (reasoning stream), SIG-POWER (engagement patterns reveal influence networks), SIG-PATH (mutual followers/engagement as weak connection signal)
Phase Recommendation	P2 — Twitter/X API access is complex and costly post-2023 changes. Low source reliability (EPIS) means the signal adds marginal confidence. Worth adding but not MVP-critical.

GOVERNANCE DECISION REQUIRED: Should Twitter/X be a new source layer feeding into existing streams (SIG-EXEC, SIG-NEWS) rather than a standalone sub-stream? The signal meaning isn't new — it's executive beliefs and market news. What's new is the source. Recommendation: add as a source layer under SIG-EXEC and SIG-NEWS with its own EPIS profile (E=0.5 P=0.9 I=0.3 S=0.4 — high proximity, low independence, low specificity), not a standalone stream.

SAFE DEFAULT (if unresolved): Engine operates without Twitter/X signals. EXEC_BELIEF_TIMELINE relies on conference speeches and interviews only — less timely, less frequent, but higher quality. No scoring degradation for accounts where exec Twitter activity is low.

OPEN-GAP-SIG-05: SIG-ANALYST — Industry Analyst Report Signals**MEDIUM****V1.3 FSD Reference:** Not referenced in V1.3 or V2.

FAILURE PREVENTED: Gartner Magic Quadrants, IDC MarketScape reports, McKinsey KSA banking studies, and BCG fintech analyses directly shape what bank CxOs prioritize and which vendors they evaluate. A Gartner MQ placing Decimal's category as 'emerging' creates a timing window; a McKinsey report on KSA digital lending creates a narrative that bank executives adopt. Without this stream, the engine cannot detect the external narrative forces that shape buying decisions.

Root Cause: Industry analyst reports are not event-driven signals in the traditional sense — they are narrative-shaping forces. The 5-Whys methodology focused on observable events (news, hires, financials), not on the meta-layer of what influences executive thinking about those events.

Candidate Source Layer	Gartner (publication monitoring), IDC, McKinsey Global Institute (KSA reports), BCG (fintech reports), Fintech Saudi publications, SAMA research papers
Signal Meaning	Narrative shaping — what the market is being told to prioritize. Influences EXEC_BELIEF_TIMELINE indirectly (executives read these reports) and directly (executives cite them in speeches/interviews).
Scoring Trigger Example	Gartner publishes 'Magic Quadrant for Digital Banking Platforms' placing Backbase as Leader -> SIG-VENDOR competitive intelligence update + EXEC_BELIEF_TIMELINE influence factor + messaging angle (Decimal vs. MQ leaders)
Cadence	On-publish (monthly scan for major reports)
Existing Section Binding	Feeds EXEC_BELIEF_TIMELINE (reasoning stream — external influence on exec thinking), SIG-VENDOR (competitive landscape validation), §7 Trust & Influence (narrative alignment — does Decimal's message align with what analysts are telling the market?)
Phase Recommendation	P2 — Human-curated input. Low automation potential due to paywalls and interpretation requirements. High value for message quality but not for scoring accuracy.

GOVERNANCE DECISION REQUIRED: Should analyst report ingestion be automated or human-curated? Analyst reports are behind paywalls (Gartner, IDC) and their signal value requires interpretation, not just detection. Recommendation: human-curated input (BD team flags relevant reports) with structured Signal entity creation, not automated scraping.

SAFE DEFAULT (if unresolved): Engine operates without analyst-report signals. Messaging and timing are not informed by the narrative forces that shape executive priorities. No scoring impact, but message relevance may be lower when bank executives are acting on analyst recommendations the engine doesn't know about.

OPEN-GAP-SIG-06: SIG-PARTNER — MOU & Partnership Signals**HIGH**

V1.3 FSD Reference: Partially covered by SIG-NEWS (partnership announcements surface as news). Not explicitly specified as a signal type with its own scoring logic.

FAILURE PREVENTED: A Saudi bank signing an MOU with a Decimal competitor (e.g., Backbase partnership with Bank X) is a high-urgency negative signal — that account's vendor evaluation may be closing. Conversely, a bank partnering with a complementary player (e.g., open banking middleware) creates an integration opportunity. MOUs and partnerships carry different signal meaning than general news: they indicate committed strategic direction, not just interest. SIG-NEWS captures the event but not the competitive/complementary classification.

Root Cause: Partnership announcements surface in SIG-NEWS as generic news items. The spec does not distinguish between 'bank announces digital initiative' (intent signal) and 'bank signs MOU with Backbase' (competitive closure signal). These require different scoring responses — one is opportunity, the other is urgency or disqualification.

Candidate Source Layer	SIG-NEWS output (reclassified), bank press releases, partner company press releases, Fintech Saudi ecosystem announcements, SAMA sandbox partner lists
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Signal Meaning	Strategic commitment — competitive closure (negative) or integration opportunity (positive). Not intent but action.
Scoring Trigger Example	Bank signs MOU with Backbase for digital banking platform -> CRITICAL competitive signal + SIG-VENDOR update (Backbase = selected vendor) + opportunity state transition (COMPETITOR_SELECTED or INTEGRATION_OPPORTUNITY depending on Decimal's relationship to partner)
Cadence	Event-driven (via SIG-NEWS reclassification) + weekly partner-ecosystem scan
Existing Section Binding	Feeds SIG-VENDOR (competitive intelligence), M2 scoring (competitive urgency modifier), Opportunity State Machine (§14 — COMPETITOR_SELECTED state), §9 relationship strategy (urgency escalation or pivot to integration angle)
Phase Recommendation	P1 — No new source integration required. Classification logic applied to SIG-NEWS output. Requires competitive-vendor registry (already partially specified in SIG-VENDOR) to classify partner announcements.

GOVERNANCE DECISION REQUIRED: Should SIG-PARTNER be a standalone sub-stream or a classification layer on SIG-NEWS output? The source is the same (news/press). What differs is the signal interpretation: SIG-NEWS treats all announcements equally; SIG-PARTNER applies competitive/complementary classification. Recommendation: classification layer on SIG-NEWS with explicit scoring rules, not a separate harvester — same source, different interpretation.

SAFE DEFAULT (if unresolved): Engine operates without partnership classification. Competitor-selection signals are treated as generic news with no urgency escalation. Accounts where a competitor has been selected continue receiving standard outreach instead of urgent intervention or graceful deprioritization.

OPEN-GAP-SIG-07: SIG-IP — Patent & Trademark Filings

LOW

V1.3 FSD Reference: Not referenced in V1.3 or V2.

FAILURE PREVENTED: Saudi bank subsidiaries and fintech ventures file patents or trademarks with SAIP (Saudi Authority for Intellectual Property) that reveal technology investment direction. A patent filing for 'AI-driven credit scoring' indicates technology build vs. buy decisions. However, patent filing rates in KSA banking are low compared to other signal sources, and the filing-to-implementation lag is long (1-3 years), reducing scoring relevance.

Root Cause: Patent monitoring is standard in deep-tech ABM but atypical for banking-sector ABM. The manual BD methodology never considered IP signals because KSA banks rarely file patents for their technology initiatives.

Candidate Source Layer	SAIP (saip.gov.sa), WIPO (international filings), Google Patents
Signal Meaning	Technology direction — build vs. buy indicator. Low confidence, long lag.
Scoring Trigger Example	Bank subsidiary files patent for 'Digital Onboarding System' -> weak signal that internal build is underway -> potential negative for Decimal (competitor is building in-house) or positive (they're investing in the space and may need platform support)
Cadence	Monthly scan
Existing Section Binding	Feeds SIG-VENDOR (build vs. buy inference), EXEC_BELIEF_TIMELINE (technology investment direction — weak signal)
Phase Recommendation	P2 or DROP — Defer unless patent filing rates in KSA fintech increase materially. Manual input only if encountered.

GOVERNANCE DECISION REQUIRED: Should SIG-IP be included at all, or is the signal-to-noise ratio too low for KSA banking? Patent activity in Saudi financial services is minimal. The resource cost of monitoring may exceed the signal value. Recommendation: defer to P2 with a manual-input option if the BD team encounters relevant filings.

SAFE DEFAULT (if unresolved): Engine operates without IP signals. No impact on scoring accuracy given the low filing rate in KSA banking. This is the lowest-priority gap.

CROSS-CUTTING CONCERNS

1. EPIS Profiles Required for All New Sources

Every new source introduced by these gaps must receive an EPIS (Evidence, Proximity, Independence, Specificity) base profile before integration, per the EPIS-ENRICH-001 pattern identified in the coherence audit (COH-F02). This is not optional — ungoverned sources degrade the Intelligence Confidence Score for every account they touch.

Source	E	P	I	S	Rationale
LinkedIn Jobs	0.8	0.7	0.8	0.9	Verified postings, high specificity (role/requirements named)
Bayt / GulfTalent	0.7	0.5	0.7	0.8	Aggregated; may lag LinkedIn by days
Etimad (tenders)	0.95	0.9	0.95	0.9	Government-official procurement; near-SAMA reliability
Crunchbase	0.8	0.5	0.8	0.7	Verified rounds; proximity low (reported, not direct)
MAGNiTT	0.75	0.6	0.75	0.8	MENA-specific; better regional coverage than Crunchbase
Twitter/X	0.5	0.9	0.3	0.4	Self-reported, unverified, high noise; proximity only value
Analyst Reports	0.85	0.3	0.9	0.6	High independence but low proximity (market-level, not account-level)
SAIP (patents)	0.9	0.4	0.9	0.5	Official filing but low proximity/specificity to buying intent

2. M0 Governance Impact

Every new source adds load to M0 (Intelligence Governance Engine). New signal types require: new dedup fingerprint patterns (a job posting and a news article about the same hire must collapse), new freshness TTL curves (job postings decay differently from news), and new source-trust weights in ICS-SOURCE-001. M0's existing rules (M0-GATE-001, M0-CONF-001, M0-RECONCILE-001) apply without modification, but the Rule_Registry must be extended with type-specific parameters for each new signal type.

3. Coherence Audit Cross-Reference

COH-F02 (Enrichment Sources Without EPIS Confidence Scores) applies directly: the EPIS profiles above are the resolution pattern. OPEN-GAP-SIG-01 (SIG-HIRE) also intersects with COH-D03 if it exists — the V1.3-to-V2 signal-type regression. The coherence audit should flag the V1.3 'Hiring' and 'Funding' references as unresolved regressions until OPEN-GAP-SIG-01 and OPEN-GAP-SIG-03 are resolved.

RECOMMENDED RESOLUTION PATH

Priority	Gap	Resolution Type	Phase	Action
1	SIG-HIRE	New 9th sub-stream	P1	Full 5-Whys + rules + source integration
2	SIG-PARTNER	Classification layer on SIG-NEWS	P1	Competitive/complementary classifier + scoring rules
3	SIG-TENDER	Manual input (P1) + automated (P2)	P1/P2	Signal entity for manual RFP input; Etimad automation later
4	SIG-FUND	New sub-stream or SIG-FIN extension	P1	MAGNiTT/Crunchbase integration
5	SIG-SOCIAL	Source layer under SIG-EXEC + SIG-NEWS	P2	Twitter/X API; low priority due to EPIS profile
6	SIG-ANALYST	Human-curated input	P2	Structured manual entry; no automation
7	SIG-IP	Defer or drop	P2/DROP	Low signal value in KSA banking context